Final Assignment Kayla Trogdon 12/10/2025

Test Cases:

X15 = 8 X16 = 4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,8
addi x16,x0,4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 51]: bge x2,x15,32
[line 52]: add x17,x17,x16
[line 53]: addi x2,x2,1
[line 54]: jal x0,-12
[line 51]: bge x2,x15,32
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	0	0x00000000	0b00000000000000000000000000000000
0	x2 (sp)	8	0x00000008	0b00000000000000000000000000001000
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	8	0x00000008	0b00000000000000000000000000001000
0	x16 (a6)	4	0x00000004	0b00000000000000000000000000000100
0	x17 (a7)	32	0x00000020	0b00000000000000000000000000100000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = 8 X16 = -4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,8
addi x16,x0,-4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 51]: bge x2,x15,32
[line 52]: add x17,x17,x16
[line 53]: addi x2,x2,1
[line 54]: jal x0,-12
[line 51]: bge x2,x15,32
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	0	0x00000000	0b00000000000000000000000000000000
0	x2 (sp)	8	0x00000008	0b00000000000000000000000000001000
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	8	0x00000008	0b00000000000000000000000000001000
0	x16 (a6)	-4	0xfffffffc	0b11111111111111111111111111111100
0	x17 (a7)	-32	0xffffffe0	0b11111111111111111111111111100000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = -8 X16 = 4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,-8
addi x16,x0,4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 43]: bge x1,x16,72
[line 44]: add x17,x17,x15
[line 45]: addi x1,x1,1
[line 46]: jal x0,-12
[line 43]: bge x1,x16,72
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	4	0x00000004	0b00000000000000000000000000000100
0	x2 (sp)	0	0x00000000	0b00000000000000000000000000000000
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	-8	0xfffffff8	0b11111111111111111111111111111000
0	x16 (a6)	4	0x00000004	0b00000000000000000000000000000100
0	x17 (a7)	-32	0xffffffe0	0b11111111111111111111111111100000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = -8 X16 = -4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,-8
addi x16,x0,-4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 55]: bge x2,x3,16
[line 56]: add x17,x17,x4
[line 57]: addi x2,x2,1
[line 58]: jal x0,-12
[line 55]: bge x2,x3,16
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	0	0x00000000	0b00000000000000000000000000000000
0	x2 (sp)	8	0x00000008	0b00000000000000000000000000001000
0	x3 (gp)	8	0x00000008	0b00000000000000000000000000001000
0	x4 (tp)	4	0x00000004	0b00000000000000000000000000000100
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	-8	0xfffffff8	0b11111111111111111111111111111000
0	x16 (a6)	-4	0xfffffffc	0b11111111111111111111111111111100
0	x17 (a7)	32	0x00000020	0b00000000000000000000000000100000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = 0 X16 = -4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,0
addi x16,x0,-4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 19]: add x17,x17,x15
[line 20]: addi x2,x2,1
[line 21]: jal x0,-12
[line 18]: bge x2,x4,16
[line 22]: jal x0,164
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	0	0x00000000	0b00000000000000000000000000000000
0	x2 (sp)	4	0x00000004	0b00000000000000000000000000000100
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	4	0x00000004	0b00000000000000000000000000000100
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	0	0x00000000	0b00000000000000000000000000000000
0	x16 (a6)	-4	0xfffffffc	0b11111111111111111111111111111100
0	x17 (a7)	0	0x00000000	0b00000000000000000000000000000000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = 8 X16 = 0:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,8
addi x16,x0,0
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 24]: add x17,x17,x16
[line 25]: addi x2,x2,1
[line 26]: jal x0,-12
[line 23]: bge x2,x15,16
[line 27]: jal x0,140
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	0	0x00000000	0b00000000000000000000000000000000
0	x2 (sp)	8	0x00000008	0b00000000000000000000000000001000
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	8	0x00000008	0b00000000000000000000000000001000
0	x16 (a6)	0	0x00000000	0b00000000000000000000000000000000
0	x17 (a7)	0	0x00000000	0b00000000000000000000000000000000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = 0 X16 = 4:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,0
addi x16,x0,4
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 34]: add x17,x17,x15
[line 35]: addi x1,x1,1
[line 36]: jal x0,-12
[line 33]: bge x1,x16,16
[line 37]: jal x0,96
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	4	0x00000004	0b00000000000000000000000000000100
0	x2 (sp)	0	0x00000000	0b00000000000000000000000000000000
0	x3 (gp)	0	0x00000000	0b00000000000000000000000000000000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	0	0x00000000	0b00000000000000000000000000000000
0	x16 (a6)	4	0x00000004	0b00000000000000000000000000000100
0	x17 (a7)	0	0x00000000	0b00000000000000000000000000000000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000

X15 = -8 X16 = 0:

Input your RISC-V code here:

```
add x1,x0,x0 #control x16 --> b
add x2,x0,x0 #controls x15 --> a
addi x15,x0,-8
addi x16,x0,0
add x17,x0,x0

beq x15,x0,12
beq x16,x0,24
jal x0,116
blt x16,x0,8
jal x0,88
sub x4,x0,x16
jal x0,20
blt x15,x0,8
jal x0,32
```

Reset | Stop | CPU: 32 Hz

```
[line 29]: add x17,x17,x16
[line 30]: addi x1,x1,1
[line 31]: jal x0,-12
[line 28]: bge x1,x3,16
[line 32]: jal x0,104
No more instructions to run! Press Reset to reload the code!
```

Features

Init Value	Register	Decimal	Hex	Binary
0	x0 (zero)	0	0x00000000	0b00000000000000000000000000000000
0	x1 (ra)	8	0x00000008	0b00000000000000000000000000001000
0	x2 (sp)	0	0x00000000	0b00000000000000000000000000000000
0	x3 (gp)	8	0x00000008	0b00000000000000000000000000001000
0	x4 (tp)	0	0x00000000	0b00000000000000000000000000000000
0	x5 (t0)	0	0x00000000	0b00000000000000000000000000000000
0	x6 (t1)	0	0x00000000	0b00000000000000000000000000000000
0	x7 (t2)	0	0x00000000	0b00000000000000000000000000000000
0	x8 (s0/fp)	0	0x00000000	0b00000000000000000000000000000000
0	x9 (s1)	0	0x00000000	0b00000000000000000000000000000000
0	x10 (a0)	0	0x00000000	0b00000000000000000000000000000000
0	x11 (a1)	0	0x00000000	0b00000000000000000000000000000000
0	x12 (a2)	0	0x00000000	0b00000000000000000000000000000000
0	x13 (a3)	0	0x00000000	0b00000000000000000000000000000000
0	x14 (a4)	0	0x00000000	0b00000000000000000000000000000000
0	x15 (a5)	-8	0xfffffff8	0b11111111111111111111111111111000
0	x16 (a6)	0	0x00000000	0b00000000000000000000000000000000
0	x17 (a7)	0	0x00000000	0b00000000000000000000000000000000
0	x18 (s2)	0	0x00000000	0b00000000000000000000000000000000
0	x19 (s3)	0	0x00000000	0b00000000000000000000000000000000
0	x20 (s4)	0	0x00000000	0b00000000000000000000000000000000